

Class XI : ARTIFICIAL INTELLIGENCE

Instructor: Saptarshi Jana

Python programming for practical Exam 2025

Assignment 4-5

Due Date: 08.09.26

20 marks

INSTRUCTIONS

- *Explain the algorithm and use variable description*
 - *Try not to use the inbuilt functions unless obvious.*
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Assignment 4

Python programming for practical Exam - I

Question 1: Write a program to swap two numbers without using a third variable.

Python Basics

Question 2: Take a string as input and count how many vowels (a, e, i, o, u) it contains.

Strings

Question 3: Check whether a given number is even or odd using conditional statements.

Conditionals

Question 4: Reverse a string using slicing.

Strings

Question 5: Given a list of numbers, find the largest and smallest number.

Lists

Question 6: Write a function that takes a number and returns its factorial.

Functions

Question 7: Check if a given year is a leap year.

Conditionals

Question 8: Print the multiplication table of a given number.

Loops

Question 9: Remove all repeated elements from a list.

Lists

Question 10: Write a function `count_char(string, ch)` that counts how many times a character appears in a string.

Functions + Strings

Question 11: Find the second largest number in a list without using any in-built functions like `max()` or `sort()`.

Lists

Question 12: Check whether a number is a strong number (sum of factorials of digits = number).

Numbers

Assignment 5

Python programming for practical Exam - II

Question 1: Accept a sentence from the user and count the frequency of each word.

Strings + Dictionaries

Question 2: Implement a function to compute the Fibonacci sequence up to n terms using recursion.

Recursion

Question 3: Write a program using a function to calculate the greatest common divisor (GCD) of two numbers.

Functions + Numbers

Question 4: Take a string input and return the longest word in the string.

Strings

Question 5: Check if two strings are anagrams of each other.

Strings

Question 6: Write a function to check whether a given number is prime.

Functions

Question 7: Sort a list of tuples based on the second element of each tuple.

Tuples + Sorting

Question 8: Create a function `calculate_grade(marks)` which accepts marks out of 100 and returns “A”, “B”, “C”, “D” or “Fail” based on ranges.

Functions + Conditionals

Question 9: Print a pyramid pattern of stars for a given n.

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* *
* * *
* * * *
* * * * *
```

Loops + Patterns

Question 10: Simulate a basic calculator performing +, −, *, and / using functions.

Functions + Operators

Question 11: Write a function that accepts a list of numbers and returns the sum of squares of all numbers.

Functions + Lists

Question 12: Define a function `is_palindrome(word)` that checks whether a word is a palindrome. Test it with different inputs.

Functions + Strings